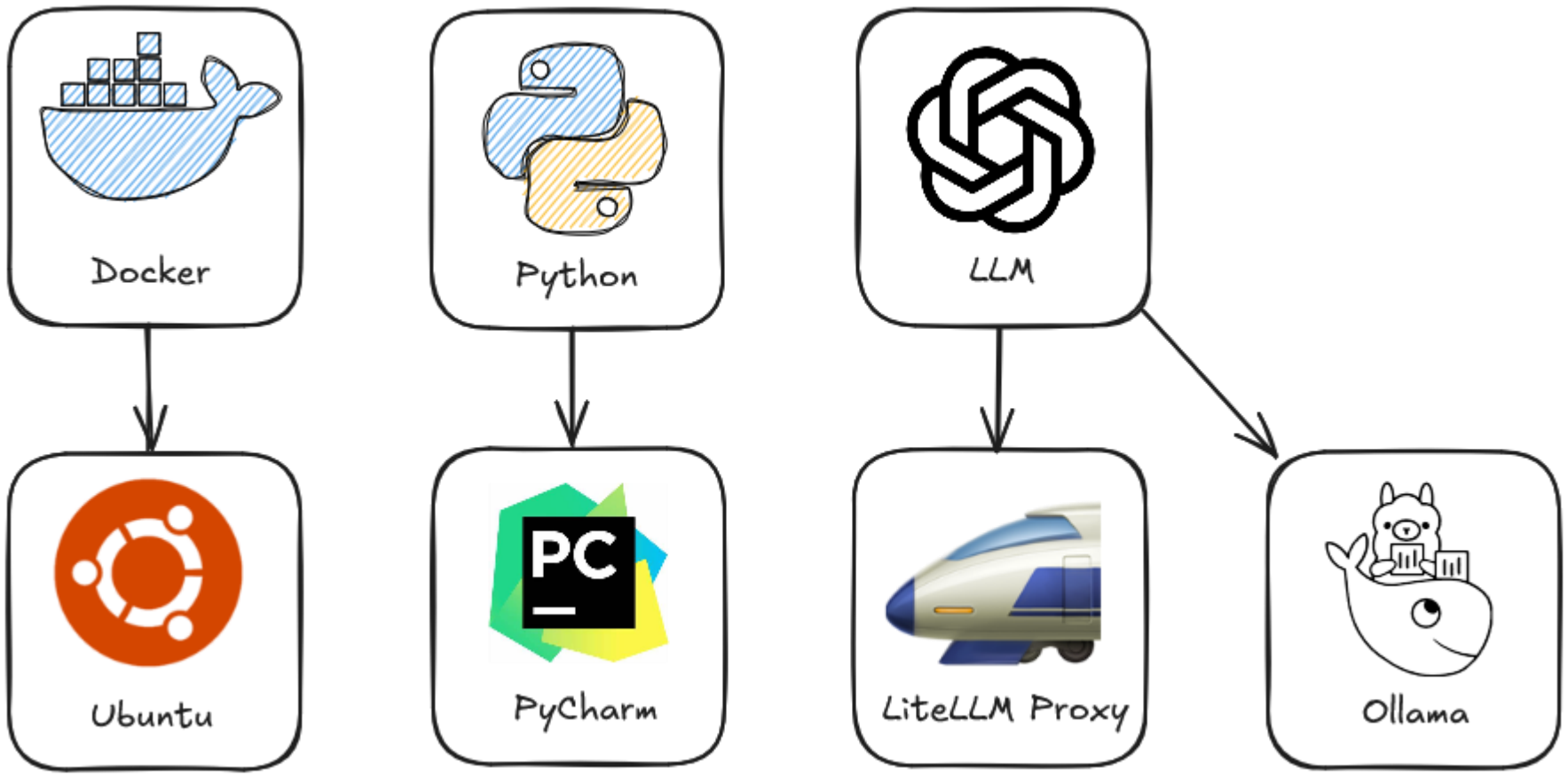
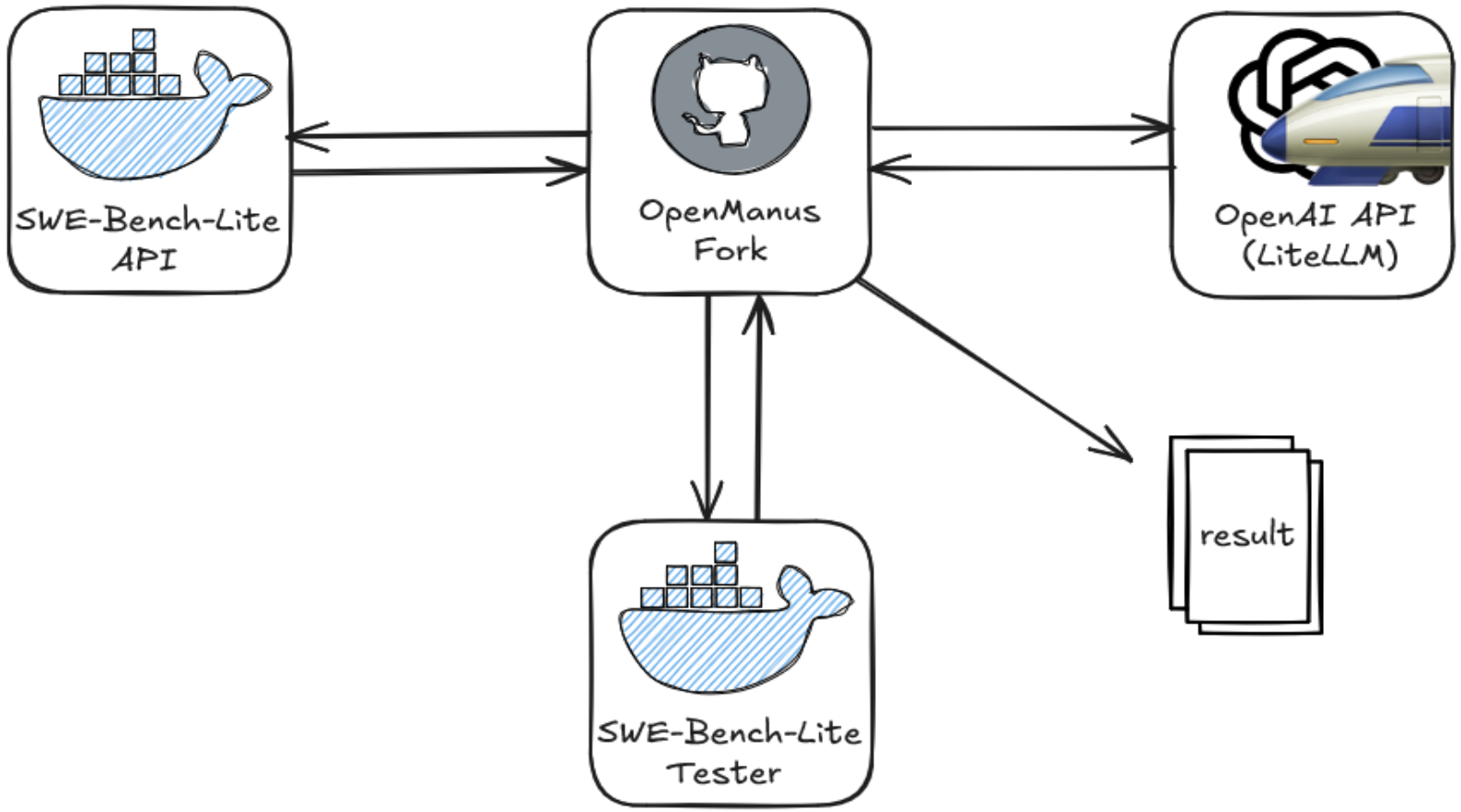
Automated Software Engineering

Phase 2: Preparation of the practical evaluation

requirements and recommendations



system overview



setup docker containers

1. install docker software
   - https://docs.docker.com/get-started/get-docker/
   - Ubuntu:
     1. sudo apt update
     2. sudo apt install docker.io
   - Windows requirements:
     → Virtualization enabled
     → WSL 2 backend
   - Windows setup options
     → use WSL 2 backend

https://www.youtube.com/watch?v=ZyBBv1JmnWQ

setup docker containers

2. run local llm with ollama
   - https://hub.docker.com/r/ollama/ollama
   - Ubuntu:
     1. sudo docker pull ollama/ollama
     2. sudo docker run -d \
        -v ollama:/root/.ollama \
        -p 11434:11434 \
        --name ollama ollama/ollama
     3. sudo docker exec -it ollama ollama run **gemma3:1b** (STRG+D)
     4. browser test: http://localhost:11434 → ollama is running

setup docker containers

3. run swe-bench-lite-api
   - https://hub.docker.com/r/paulroewer/swe-bench-lite-api
   - https://github.com/PaulRoewer/swe-bench-lite-api
   - Ubuntu:
     1. sudo docker pull paulroewer/swe-bench-lite-api
     2. sudo docker run -d \
        -p 8081:8080 \
        --name swe-bench-lite-api \
        paulroewer/swe-bench-lite-api

setup docker containers

4. run swe-bench-lite-tester
   - https://hub.docker.com/r/paulroewer/swe-bench-lite-tester
   - https://github.com/PaulRoewer/swe-bench-lite-tester
   - Ubuntu:
     1. sudo docker pull paulroewer/swe-bench-lite-tester
     2. sudo docker run -d \
        -p 8082:8080 \
        --name swe-bench-lite-tester \
        -v **/mnt/d/workspace/OpenManus/workspace/repos:/repos** \
        -v /var/run/docker.sock:/var/run/docker.sock \
        paulroewer/swe-bench-lite-tester

LLM configurations

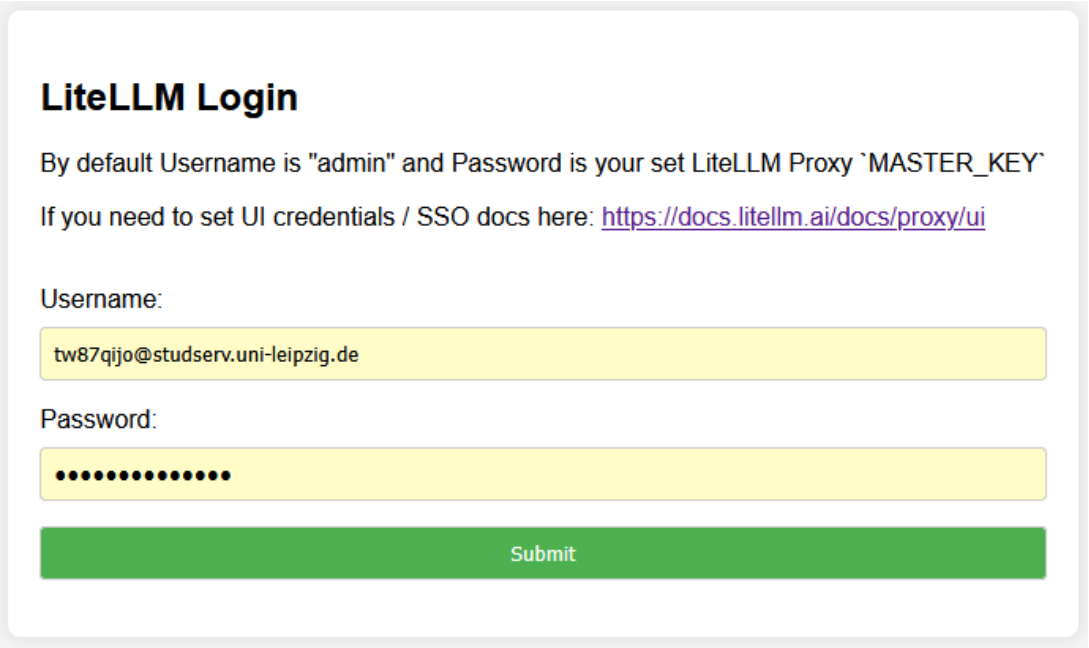
OLLAMA

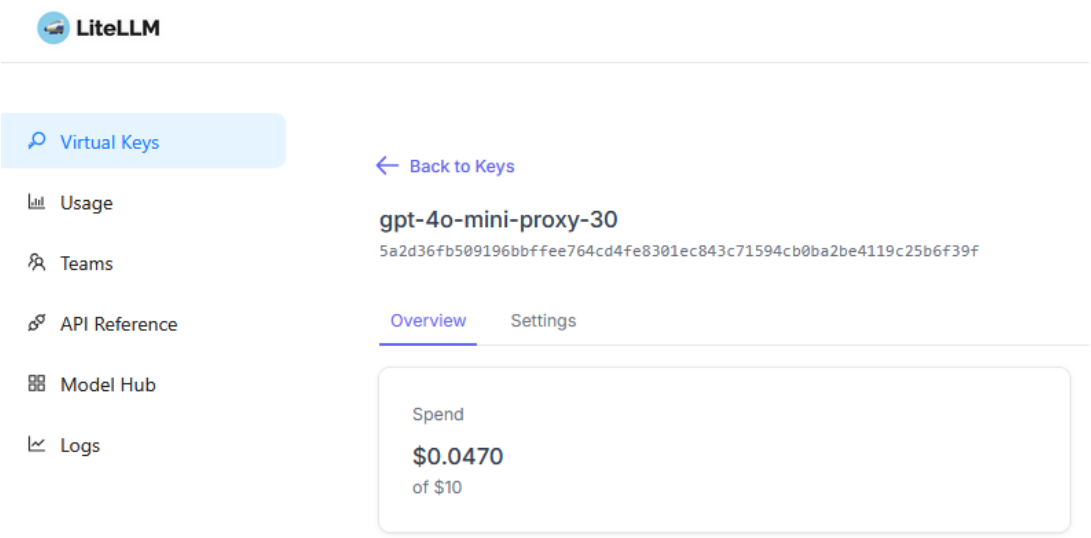
- api_type = 'ollama'
- model = "gemma3:1b"
- base_url = "http://localhost:11434/v1"
- api_key = "ollama"
- max_tokens = 4096
- temperature = 0.0

OPENAI (LiteLLM Proxy)

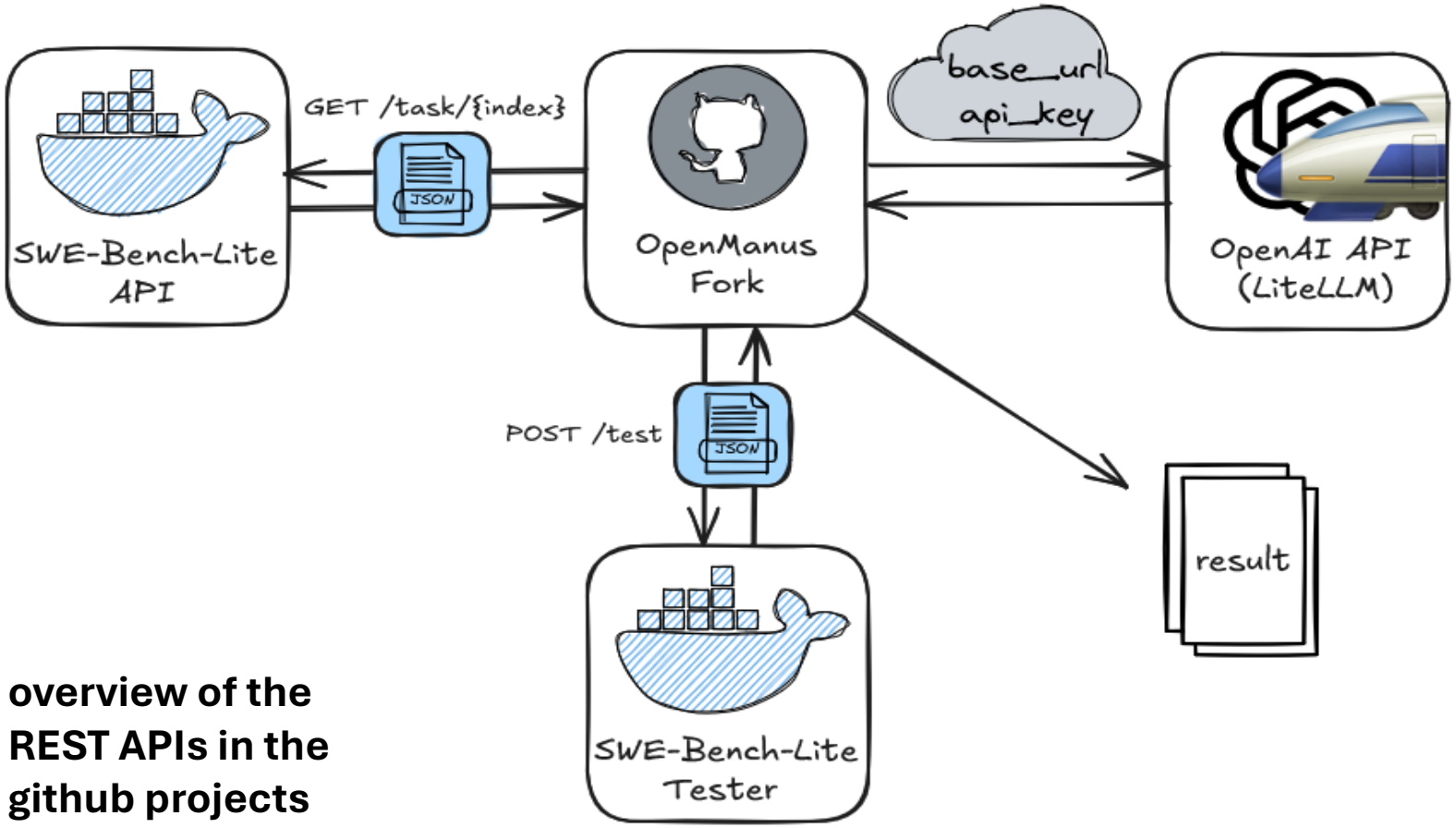
- api_type= 'openai'
- model = "gpt-4o-mini"
- base_url = "http://188.245.32.59:4000/v1"
- api_key = "sk-..."
- max_tokens = 8096
- temperature = 0.0

LiteLLM API → http://188.245.32.59:4000/ui/



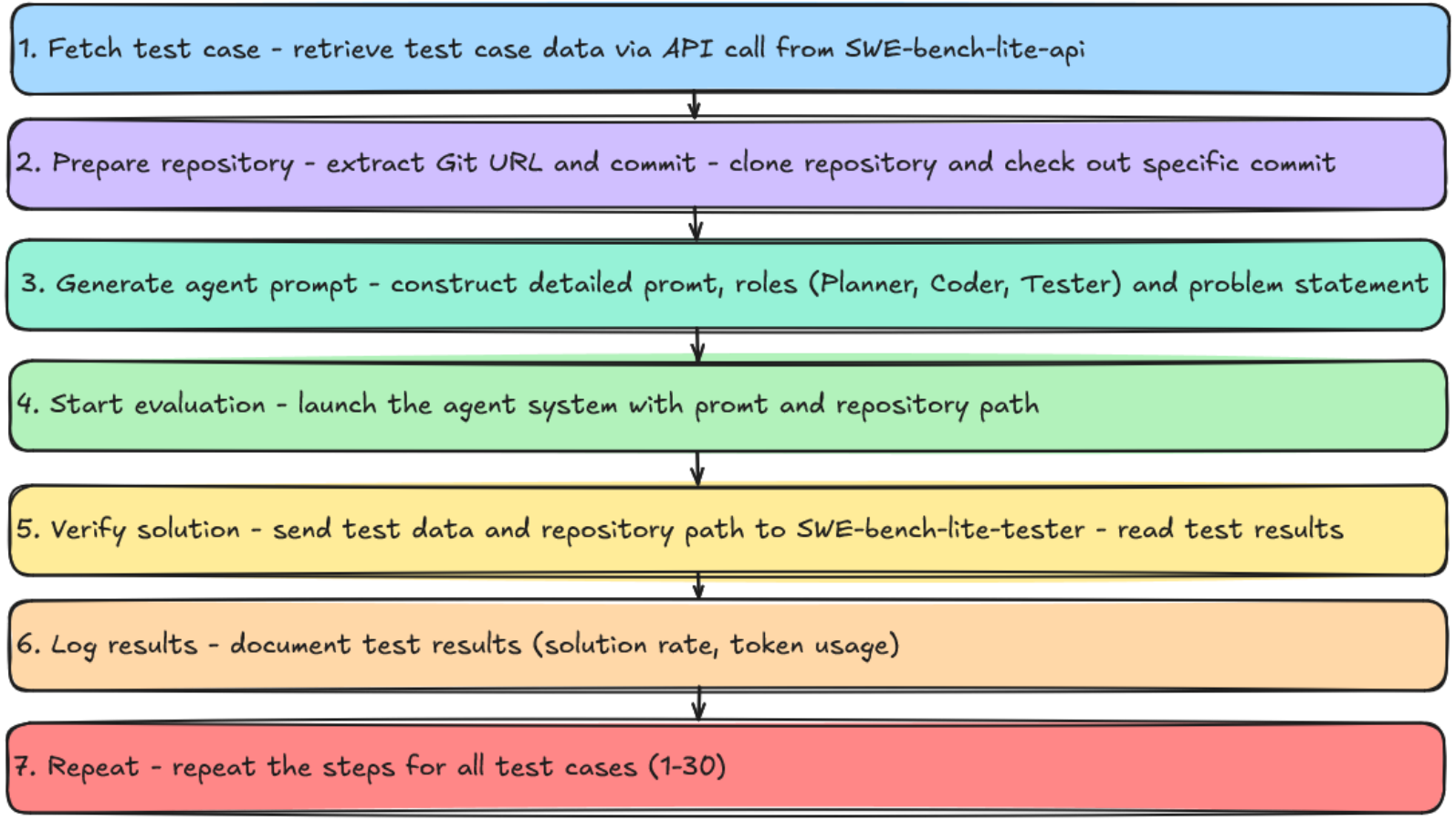


system overview



overview of the REST APIs in the github projects

agent in action: workflow



live demo

OpenManus

https://github.com/PaulRoewer/OpenManus

advice

"The results are of secondary importance;
the main focus lies in the process that leads to them."

SWE-bench Analysis

Home | Paper | Code | Submit

Select the split & model below to get automated analyses of the model's performance on the SWE-bench split.

Split: Lite Model: (2024-07-28) SWE-agent + GPT 4o (2024-05-13) Load Data

Viewing 20240728_sweagent_gpt4o's performance on the SWE-bench lite split, which resolved 18.33% of 300 issues. (Logs)

SWE-agent

SWE-agent turns LMs (e.g. GPT-4) into software engineering agents that can resolve issues in real GitHub repositories.

Preprint | GitHub | Website | Documentation | Discord

We accomplish our results by designing simple LM-centric commands and feedback formats to make it easier for the LM to browse the repository, view, edit and execute code files. We call this an **Agent-Computer Interface (ACI)**.

SWE-agent is built and maintained by researchers from Princeton University.

If you found this work helpful, please consider citing it using the following:

```
@misc{yang2024sweagent,
      title={SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering},
      author={John Yang and Carlos E. Jimenez and Alexander Wettig and Kilian Lieret and Shunyu Yao and Karthik Narasimhan and Ofir
Press},
      year={2024},
      eprint={2405.15793},
      archivePrefix={arXiv},
      primaryClass={cs.SE}
}
```

% Resolved by Repository

Repository	Resolved	Total	% Resolved
astropy/astropy	1	6	16.67%
django/django	26	114	22.81%
matplotlib/matplotlib	0	23	0.00%
mwaskom/seaborn	2	4	50.00%
pallets/flask	0	3	0.00%
psf/requests	2	6	33.33%
pydata/xarray	1	5	20.00%
pylint-dev/pylint	0	6	0.00%
pytest-dev/pytest	5	17	29.41%
scikit-learn/scikit-learn	6	23	26.09%
sphinx-doc/sphinx	0	16	0.00%
sympy/sympy	12	77	15.58%

https://www.swebench.com/viewer.html

need support?

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